HSAN ENNOURI

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EE Master's student at EPFL. Seeking a 6-month Master's thesis internship from July/September 2027 in hardware design or biomedical engineering.

SKILLS

- **Programming**
  C++, Python, MATLAB, Assembly
- **Circuit simulation**
  LTspice, PLECS
- **Electronics CAD**
  KiCad, Altium
- **Mechanical CAD**
  CATIA
- **Digital Logic**
  Logisim evolution
- **Tools**
  Git, Linux, Overleaf, STM32 IDE

EDUCATION

- **M.Sc. Electrical & Electronic Engineering — EPFL**
  2025 - Present
- **B.Sc. Electrical & Electronic Engineering — EPFL (GPA: 5.73/6)**
  2022 - 2025
- **Exchange — NUS**
  Aug 2024 - May 2025
- **Cours de Mathématiques spéciales: CMS — EPFL**
  2021 - 2022

LANGUAGES

- **French:** Fluent
- **English:** Fluent
- **Arabic:** Native

INTERESTS

- **Sports** Tennis, Table Tennis, Volleyball
- **Dancing** Salsa, Jive, Bachata
- **Martial Arts** Kick Boxing, Muay Thai
- **Photography**
- **Cooking**

HONORS & AWARDS

EPFL Master Excellence Fellowship

September 2025 – Present

- Competitive scholarship awarded to Master's students who achieve outstanding academic results and distinguish themselves through extracurricular engagement.

PROFESSIONAL ACTIVITIES

Machine Learning Internship (at EOSS SA)

June 2026 – August 2026 (ongoing) | Morges, Switzerland

- Developing machine learning methods for signal classification on distributed sensing systems, including feature engineering and model evaluation for a real-time, resource-constrained classifier.

Summer In the Lab Internship

Microbiorobotic Systems Laboratory (MICROBS)

July 2025 – August 2025 | EPFL, Switzerland

- Redesigned the readout circuit architecture for the **CoroSense** multi-sensor endovascular guidewire, targeting real-time blood flow monitoring across 4 simultaneous pressure acquisition channels.
- Migrated control logic from **Arduino** to **STM32**, leveraging higher clock speeds, expanded memory and advanced peripheral capabilities for reliable multi-channel acquisition.
- Designed a custom PCB integrating instrumentation amplifiers (AD623) and 8th-order switched-capacitor filters (MAX7403) to condition μV-range sensor outputs

Teaching Assistant (EPFL)

2022 – Present

- Conducted exercise and lab sessions and provided student support across 10 different courses: Analysis I - IV, Electronics I / II, Logic systems, Electrical engineering science and technology, Electrotechnics and Microcontrollers.

CO-CURRICULAR ACTIVITIES

Make Projects EPFL

EPFL Xplore - ERC Electronics Team

September 2025 – Present

- Redesigned the navigation board, reducing its footprint from 102 mm $\times$ 92 mm to 67 mm $\times$ 62.5 mm by replacing the EPOS4 Compact 50/8 CAN motor controller with the EPOS4 Micro 24/5 CAN and switching from XT60 to XT30 connectors.
- Tested the power protection board, verifying correct operation of protection circuits under nominal and fault conditions.
- Developed firmware to interface, calibrate and validate the **INA239** digital power monitor, ensuring accurate current and voltage measurements across the rover's power distribution system.

EPFL Spacecraft Team

September 2023 – June 2024

- **EPS member:** Assembled and tested the EPS PCB; authored technical documentation for onboard subsystems and the umbilical cord system.
- **Ground segment:** Contributed to mechanical integration and hardware troubleshooting of the UHF antenna ground station.

EPFL Associations & Volunteering

- **Fréquence Banane (Student Radio):** Produced and presented live broadcasts; conducted research and wrote editorial content for recurring segments; managed on-air technical setup.
- **180°C Cooking Association:** Event & communication member. Co-coordinated events including Vivapoly and the Spice League cooking contest; managed social media content.
- **Castor Freegan:** Cooked vegetarian meals from unsold produce and distributed them freely to students to reduce food waste.
- **CMS Coach:** Guided first-year preparatory students through campus integration, academic adaptation and social life through organized events.